



Sydney Girls High School

2010  
TRIAL HIGHER SCHOOL  
CERTIFICATE EXAMINATION

# Mathematics

## Extension 1

This is a trial paper ONLY. It does not necessarily reflect the format or the content of the 2010 HSC Examination Paper in this subject.

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Candidate Number

### General Instructions

- Reading Time - 5 minutes
- Working time - 2 hours
- Attempt ALL questions
- ALL questions are of equal value
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Standard integrals are supplied
- Board-approved calculators may be used.
- Diagrams are not to scale
- Each question attempted should be started on a new page. Write on one side of the paper only.

### TABLE OF STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

Note:  $\ln x = \log_e x, \quad x > 0$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left( x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left( x + \sqrt{x^2 + a^2} \right)$$

**QUESTION 1:** (12 marks)

a) Evaluate  $\int_0^2 \frac{1}{x^2+4} dx$ . 2

b) Find the acute angle between the lines  $2x - y - 1 = 0$  and  $3x - y - 2 = 0$ . Give your answer to the nearest minute if necessary. 2

c) A is the point  $(-1, 1)$  and B is the point  $(3, 3)$ . Find the point C which divides AB externally in the ratio 3 : 1. 2

d) Evaluate  $\int_{-1}^0 x\sqrt{1+x} dx$  using the substitution  $u = 1 + x$ . 3

e) Solve the inequality  $\frac{2}{x-1} \leq 1$ . 3

**QUESTION 2:** (12 marks)

a) If  $\alpha, \beta$  and  $\gamma$  are the roots of the equation  $x^3 - 2x + 5 = 0$ , find the values of:

(i)  $\alpha + \beta + \gamma$  1

(ii)  $\alpha\beta + \alpha\gamma + \beta\gamma$  1

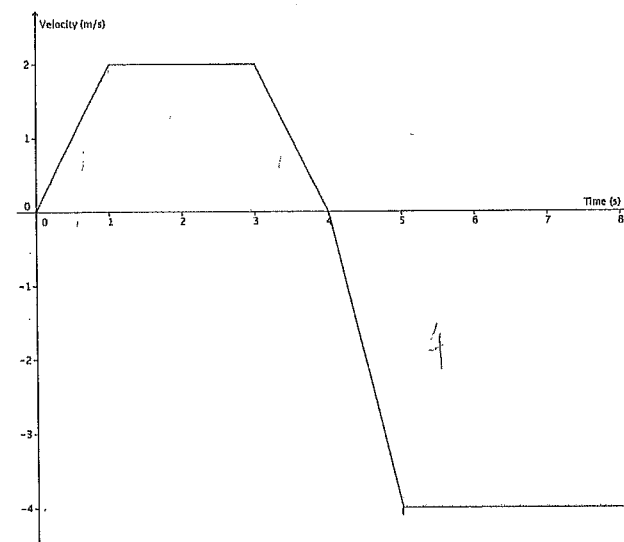
(iii)  $\alpha\beta\gamma$  1

(iv)  $\alpha^2 + \beta^2 + \gamma^2$  2

(v)  $(\alpha - 2)(\beta - 2)(\gamma - 2)$  2

b) Use mathematical induction to prove that  $n^3 + 2n$  is divisible by 3 for all  $n \geq 1$ . 3

c) The graph below shows the velocity of a particle as it moves along a straight line. Initially the particle is at the origin. At what time does the particle return to the origin? 2



**QUESTION 3:** (12 marks)

- a) Points  $P(2ap, ap^2)$  and  $Q(2aq, aq^2)$  lie on the parabola  $x^2 = 4ay$ .  $PQ$  subtends a right angle at the vertex. The tangents at  $P$  and  $Q$  meet at  $T$ .

(i) Show that  $pq = -4$ .

2

(ii) Find the equation of the locus of  $T$ .

3

b)

(i) Express  $\frac{3}{2}\cos\theta + 2\sin\theta$  in the form  $A\cos(\theta - \alpha)$  where  $A > 0$ .

2

(ii) Hence solve the equation  $3\cos\theta + 4\sin\theta = 2$  for  $0 \leq \theta \leq 360^\circ$ .

2

- c) A polynomial is given by  $P(x) = x^3 + ax^2 + bx - 18$ . Find values for  $a$  and  $b$  if  $(x+2)$  is a factor of  $P(x)$  and  $-24$  is the remainder when  $P(x)$  is divided by  $(x-1)$ .

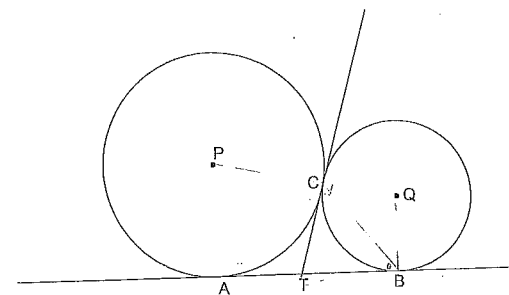
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**QUESTION 4:** (12 marks)

- a) Use Newton's method to find a second approximation to the positive root of  $x - 2\sin x = 0$ . Take  $x = 1.7$  as the first approximation.

2

- b) Two circles touch externally at  $C$ . The circles, which have centres  $P$  and  $Q$  are touched by a common tangent at  $A$  and  $B$  respectively. The common tangent through  $C$  meets the common tangent  $AB$  at  $T$ .



(i) Show that  $AT = TB$ .

2

(ii) Show that  $\angle ACB$  is a right angle.

2

c)

(i) Differentiate  $y = \tan^{-1}x + \tan^{-1}\frac{1}{x}$

3

(ii) Hence or otherwise sketch the graph of  $y = \tan^{-1}x + \tan^{-1}\frac{1}{x}$ .

3

QUESTION 4 OVER PAGE

QUESTION 5 OVER PAGE

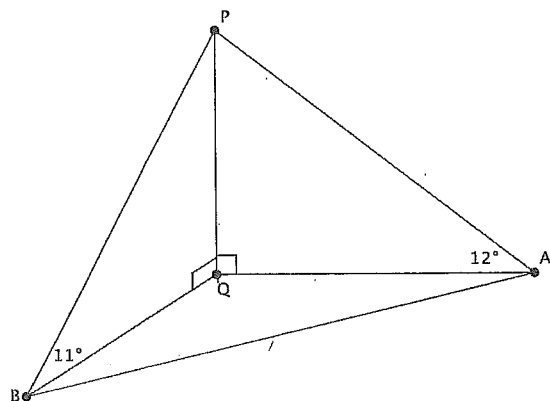
**QUESTION 5:** (12 marks)

a) Evaluate  $\lim_{x \rightarrow 0} \frac{\sin 3x}{5x}$ .

2

- b) The angle of elevation of a tower  $PQ$  of height  $h$  metres from a point  $A$ , due east of it is  $12^\circ$ . From another point  $B$ , due south of the tower, the angle of elevation is  $11^\circ$ . The points  $A$  and  $B$  are 100 m apart and on the same level as  $Q$ , the base of the tower. Calculate  $h$  to the nearest metre.

4



c) Prove, without the use of a calculator:  $\sin^{-1} \frac{1}{3} + \cos^{-1} \frac{2}{3} = \sin^{-1} \frac{2(1+\sqrt{10})}{9}$ .

3

d) Prove that  $\frac{\sin 3\theta}{\sin \theta} - \frac{\cos 3\theta}{\cos \theta} = 2$  (for  $\sin \theta \neq 0, \cos \theta \neq 0$ ).

3

**QUESTION 6:** (12 marks)

- a) Consider the equation  $x^3 + 6x^2 - x - 30 = 0$ . One of the roots of the equation is equal to the sum of the other two roots. Find the value of the three roots.

4

b) Find  $2 \int \cos^2 4x \, dx$ .

2

- c) Callum has baked a chocolate cake. At 2 p.m, he takes it out of a  $180^\circ\text{C}$  hot oven and places it on a cooling rack in the kitchen, where the temperature is  $20^\circ\text{C}$ . According to Newton's Law of Cooling the temperature  $T$ , of Callum's cake  $t$  minutes after it comes out of the oven satisfies the equation:

$$\frac{dT}{dt} = -k(T - 20) \text{ where } k \text{ is a constant.}$$

- (i) Show that  $T = 20 + 160e^{-kt}$  is a solution of the equation.

2

- (ii) At 2:15 p.m, the cake's temperature is  $100^\circ\text{C}$ . Find the value of  $k$ , correct to 3 significant figures.

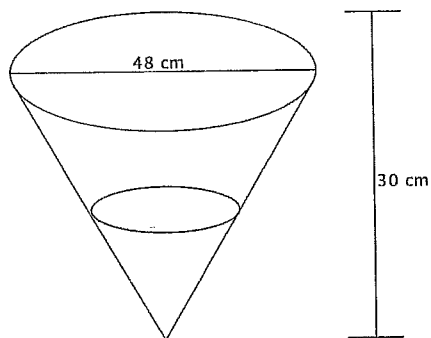
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- (iii) The cake must cool to  $35^\circ\text{C}$  before Callum can ice it. What is the earliest time that the cake can be iced?

2

**QUESTION 7:** (12 marks)

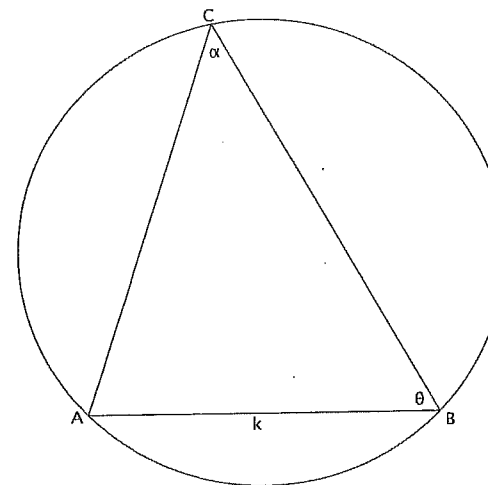
- a) Water is pouring into a conical vessel of height 30 cm and diameter 48 cm.



- (i) Show that when there is water in the vessel to a depth of  $h$  cm, the volume of water ( $V \text{ cm}^3$ ) is given by  $V = \frac{16\pi h^3}{75}$ . 2

- (ii) If the height of water is increasing at 0.5 cm/min, find the rate of increase of the volume when the depth of water is 10 cm from the top. Give your answer in exact form. 3

- b) Points  $A$ ,  $B$  and  $C$  lie on a circle. The length of the chord  $AB$  is a constant  $k$ . The radian measures of  $\angle ABC$  and  $\angle BCA$  are  $\theta$  and  $\alpha$  respectively.



- (i) Let  $l$  equal the sum of the lengths of chords  $CA$  and  $CB$ . Show that  $l = \frac{k}{\sin \alpha} [\sin \theta + \sin(\theta + \alpha)]$ . 3
- (ii) Why is  $\alpha$  a constant? 1
- (iii) Evaluate  $\frac{dl}{d\theta}$  when  $\theta = \frac{\pi}{2} - \frac{\alpha}{2}$ . 3

**END OF TEST ©**

# TRIAL HSC EXTENSION 1 MATHEMATICS 2010 - SOLUTIONS

## Question 1:

$$\begin{aligned} \text{a)} \quad \int_0^2 \frac{1}{x^2+4} dx &= \frac{1}{2} \left[ \tan^{-1} \left( \frac{x}{2} \right) \right]_0^2 \\ &= \frac{1}{2} (\tan^{-1} 1 - \tan^{-1} 0) \\ &= \frac{1}{2} \cdot \frac{\pi}{4} \\ &= \frac{\pi}{8} \end{aligned}$$

$$\begin{aligned} \text{b)} \quad 2x - y - 1 &= 0 & 3x - y - 2 &= 0 \\ y &= 2x - 1 & y &= 3x - 2 \\ m_1 &= 2 & m_2 &= 3 \end{aligned}$$

$$\begin{aligned} \tan \theta &= \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| \\ &= \left| \frac{2 - 3}{1 + 2 \times 3} \right| \\ &= \left| \frac{-1}{7} \right| \\ &= \frac{1}{7} \\ \theta &= \tan^{-1} \frac{1}{7} \\ &= 8.130102\dots \\ &= 8^\circ 8' \text{ (nearest minute)} \end{aligned}$$

$$\begin{aligned} \text{c)} \quad m:n &= -3:1 \\ x &= \frac{mx_2 + nx_1}{m+n} & y &= \frac{my_2 + ny_1}{m+n} \\ &= \frac{(-3 \times 3) + (1 \times -1)}{-3+1} & &= \frac{(-3 \times 3) + (1 \times 1)}{-3+1} \\ &= 5 & &= 4 \\ \therefore D &= (5, 4) \end{aligned}$$

$$\begin{array}{lll} \text{d)} \quad u = 1+x & \text{When } x = -1: & \text{When } x = 0: \\ \frac{du}{dx} = 1 & u = 1-1 & u = 1+0 \\ du = dx & = 0 & = 1 \end{array}$$

$$\begin{aligned} \int_{-1}^0 x\sqrt{1+x} dx &= \int_0^1 (u-1)\sqrt{u} du \\ &= \int_0^1 (u-1)u^{\frac{1}{2}} du \\ &= \int_0^1 \left( u^{\frac{3}{2}} - u^{\frac{1}{2}} \right) du \\ &= \left[ \frac{2u^{\frac{5}{2}}}{5} - \frac{2u^{\frac{3}{2}}}{3} \right]_0^1 \\ &= \left[ \frac{2\sqrt{u^5}}{5} - \frac{2\sqrt{u^3}}{3} \right]_0^1 \\ &= \frac{2}{5} - \frac{2}{3} \\ &= -\frac{4}{15} \end{aligned}$$

$$\begin{aligned} \text{e)} \quad \frac{2}{x-1} &\leq 1 \\ 2(x-1) &\leq (x-1)^2 \\ (x-1)^2 - 2(x-1) &\geq 0 \\ (x-1)[(x-1)-2] &\geq 0 \\ (x-1)(x-3) &\geq 0 \\ \therefore x &< 1 \text{ or } x \geq 3 \end{aligned}$$

Question 2:

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                      |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| a) | i.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | $\alpha + \beta + \gamma = 0$                                                                                                                                                                                                                                                                                                                        |
|    | ii.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | $\alpha\beta + \alpha\gamma + \beta\gamma = -2$                                                                                                                                                                                                                                                                                                      |
|    | iii.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | $\alpha\beta\gamma = -5$                                                                                                                                                                                                                                                                                                                             |
|    | iv.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$ $= 0^2 - 2(-2)$ $= 4$                                                                                                                                                                                                                    |
|    | v.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | $(\alpha - 2)(\beta - 2)(\gamma - 2) = (\alpha\beta - 2\alpha - 2\beta + 4)(\gamma - 2)$ $= \alpha\beta\gamma - 2\alpha\gamma - 2\beta\gamma + 4\gamma - 2\alpha\beta + 4\alpha + 4\beta - 8$ $= \alpha\beta\gamma - 2(\alpha\gamma + \beta\gamma + \alpha\beta) + 4(\alpha + \beta + \gamma) - 8$ $= (-5) - 2(-2) + 4(0) - 8$ $= -5 + 4 - 8$ $= -9$ |
| b) | <p><b>Step 1:</b> Prove true for <math>n = 1</math>:</p> $1^3 + 2(1) = 3$ <p>which is divisible by 3.<br/> <math>\therefore</math> true for <math>n = 1</math></p> <p><b>Step 2:</b> Assume true for <math>n = k</math>:</p> <p>Let <math>k^3 + 2k = 3M</math> where <math>M</math> is a positive integer.</p> <p><b>Step 3:</b> Prove true for <math>n = k + 1</math>:</p> $(k + 1)^3 + 2(k + 1) = k^3 + 3k^2 + 3k + 1 + 2k + 2$ $= 3M + 3k^2 + 3k + 3 \text{ (using assumption)}$ $= 3(M + k^2 + k + 1)$ <p>which is divisible by 3.</p> <p>Proven true for <math>n = k + 1</math></p> <p><b>Step 4:</b> If true for <math>n = k</math>, then proven true for <math>n = k + 1</math>. Proven true for <math>n = 1</math>, hence proven true for <math>n = 2</math>. If true for <math>n = 2</math>, then proven true for <math>n = 3</math> and so on. Hence, by mathematical induction, statement is true for all integers <math>n \geq 1</math>.</p> |                                                                                                                                                                                                                                                                                                                                                      |
| c) | $t = 6$ seconds.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                      |

Question 3:

|    |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                              |
|----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| a) | i.  | $m_{po} = \frac{ap^2 - 0}{2ap - 0}$ $= \frac{p}{2}$                                                                                                                                                                                                                                                                                                                                                                                                                                 | $m_{qo} = \frac{aq^2 - 0}{2aq - 0}$ $= \frac{q}{2}$                                                                                          |
|    |     | <p>As <math>OP \perp OQ</math>, <math>m_{op} \times m_{oq} = -1</math>:</p> $\frac{p}{2} \times \frac{q}{2} = -1$ $pq = -4$                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                              |
|    | ii. | <p>At <math>P</math>: <math>m_r = p</math></p> $y - ap^2 = p(x - 2ap)$ $y = px - ap^2$ <p>Eqn tangent at <math>P</math>: <math>y = px - ap^2</math></p> <p>Eqn tangent at <math>Q</math>: <math>y = qx - aq^2</math></p> <p>At <math>T</math>:</p> $px - ap^2 = qx - aq^2$ $px - qx = ap^2 - aq^2$ $x(p - q) = a(p - q)(p + q)$ $x = a(p + q)$ <p>When <math>x = a(p + q)</math>:</p> $y = p(a[p + q]) - ap^2$ $= apq$ $= -4a$ <p>Locus of <math>T</math>: <math>y = -4a</math></p> |                                                                                                                                              |
| b) | i.  | $A = \sqrt{\left(\frac{3}{2}\right)^2 + 2^2}$ $= \sqrt{\frac{9}{4} + 4}$ $= \frac{5}{2}$                                                                                                                                                                                                                                                                                                                                                                                            | $\tan \theta = \frac{2}{\left(\frac{3}{2}\right)}$ $= \frac{4}{3}$ $\theta = \tan^{-1} \frac{4}{3}$ $= 53^\circ 8' \text{ (nearest minute)}$ |
|    |     | $\frac{3}{2} \cos \theta + 2 \sin \theta = \frac{5}{2} \cos(\theta - 53^\circ 8')$                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                              |

ii.  $3\cos\theta + 4\sin\theta = 2$

$$\frac{3}{2}\cos\theta + 2\sin\theta = 1$$

$$\frac{5}{2}\cos(\theta - 53^\circ 8') = 1$$

$$\cos(\theta - 53^\circ 8') = \frac{2}{5}$$

$$\theta - 53^\circ 8' = 66^\circ 25'$$

$$\theta = 119^\circ 33'$$

$$\theta - 53^\circ 8' = 293^\circ 35'$$

$$\theta = 346^\circ 42'$$

c)  $P(x) = x^3 + ax^2 + bx - 18$

$$P(-2) = (-2)^3 + a(-2)^2 + b(-2) - 18$$

$$0 = -8 + 4a - 2b - 18$$

$$4a - 2b = 26$$

$$2a - b = 13$$

$$2a - b = 13$$

$$a + b = -7$$

$$3a = 6$$

$$a = 2$$

$$b = -9$$

$$P(1) = (1)^3 + a(1)^2 + b(1) - 18$$

$$-24 = 1 + a + b - 18$$

$$a + b = -7$$

#### Question 4:

a)  $f(x) = x - 2\sin x$

$$f(1.7) = 1.7 - 2\sin 1.7$$

$$= -0.2833296...$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$= 1.7 - \frac{f(1.7)}{f'(1.7)}$$

$$\approx 1.9 \text{ (to 1 dec. pl)}$$

$$f'(x) = 1 - 2\cos x$$

$$f'(1.7) = 1 - 2\cos(1.7)$$

$$= 1.257688989...$$

b) i.  $AT = TC$  (equal tangents from an external point)

$$TB = TC \text{ (equal tangents from an external point)}$$

$$\therefore AT = TB$$

ii. As  $AT = TC = TB$ :

$T$  is the centre of a circle with diameter  $AB$ .

$$\therefore \angle ACB = 90^\circ \text{ (angle in a semi-circle)}$$

c) i.  $y = \tan^{-1} x + \tan^{-1} \frac{1}{x}$

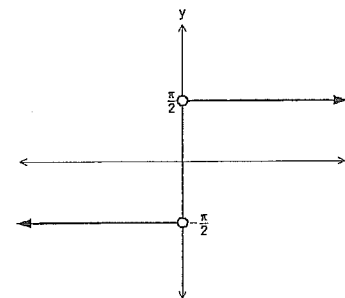
$$= \tan^{-1} x + \tan^{-1} x^{-1}$$

$$= \frac{1}{1+x^2} - \frac{1}{x^2} \left( \frac{1}{1+\frac{1}{x^2}} \right)$$

$$= \frac{1}{1+x^2} - \frac{1}{x^2+1}$$

$$= 0$$

ii.



Question 5:

$$\begin{aligned} \text{a) } \lim_{x \rightarrow 0} \frac{\sin 3x}{5x} &= \frac{3}{5} \lim_{x \rightarrow 0} \frac{\sin 3x}{3x} \\ &= \frac{3}{5} \left( \text{as } \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 \right) \end{aligned}$$

b) In  $\triangle BPQ$ :

$$\tan 11^\circ = \frac{h}{BQ}$$

$$BQ = h \cot 11^\circ$$

$$AB^2 = BQ^2 + AQ^2$$

$$100^2 = (h \cot 11^\circ)^2 + (h \cot 12^\circ)^2$$

$$= h^2 (\cot^2 11^\circ + \cot^2 12^\circ)$$

$$h^2 = \frac{100^2}{\cot^2 11^\circ + \cot^2 12^\circ}$$

$$h = 14.344 \dots \text{ m}$$

$$\approx 14 \text{ m (to the nearest metre)}$$

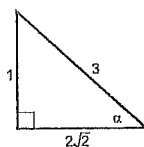
In  $\triangle APQ$ :

$$\tan 12^\circ = \frac{h}{AQ}$$

$$AQ = h \cot 12^\circ$$

$$\text{c) Let } \alpha = \sin^{-1} \frac{1}{3}$$

$$\sin \alpha = \frac{1}{3}$$



$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

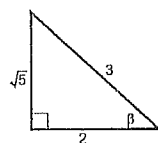
$$= \frac{1}{3} \times \frac{2}{3} + \frac{2\sqrt{2}}{3} \times \frac{\sqrt{5}}{3}$$

$$\alpha + \beta = \sin^{-1} \left( \frac{2}{9} + \frac{2\sqrt{10}}{9} \right)$$

$$\sin^{-1} \frac{1}{3} + \cos^{-1} \frac{2}{3} = \sin^{-1} \frac{2(1 + \sqrt{10})}{9}$$

$$\text{Let } \beta = \cos^{-1} \frac{2}{3}$$

$$\cos \beta = \frac{2}{3}$$



$$\begin{aligned} \text{d) } LHS &= \frac{\sin 3\theta}{\sin \theta} - \frac{\cos 3\theta}{\cos \theta} \\ &= \frac{\sin 3\theta \cos \theta - \cos 3\theta \sin \theta}{\sin \theta \cos \theta} \\ &= \frac{\sin(3\theta - \theta)}{\sin \theta \cos \theta} \\ &= \frac{\sin 2\theta}{\sin \theta \cos \theta} \\ &= \frac{2 \sin \theta \cos \theta}{\sin \theta \cos \theta} \\ &= 2 \\ LHS &= RHS \end{aligned}$$

Question 6:

a)  $x^3 + 6x^2 - x - 30 = 0$

Let roots be  $\alpha, \beta, \alpha + \beta$

$$2\alpha + 2\beta = -6$$

$$\alpha + \beta = -3 \text{ --- (1) ---}$$

$$\alpha\beta(\alpha + \beta) = 30$$

$$\alpha\beta(-3) = 30$$

$$\alpha\beta = -10$$

$$\beta = \frac{-10}{\alpha} \text{ --- (2) ---}$$

Sub (2) into (1):

$$\alpha - \frac{10}{\alpha} = -3$$

$$\alpha^2 + 3\alpha - 10 = 0$$

$$(\alpha + 5)(\alpha - 2) = 0$$

$$\alpha = -5 \text{ or } \alpha = 2$$

When  $\alpha = -5$ :  $\beta = 2$

When  $\alpha = 2$ :  $\beta = -5$

$\therefore$  The roots are -5, 2 and -3.

b)  $\cos 2x = 2\cos^2 x - 1$

$$\cos 8x = 2\cos^2 4x - 1$$

$$2\cos^2 4x = 1 + \cos 8x$$

$$2 \int \cos^2 4x dx = \int (1 + \cos 8x) dx$$

$$= x + \frac{1}{8} \sin 8x + C$$

c) i.  $T = 20 + 160e^{-kt}$

$$\frac{dT}{dt} = -160ke^{-kt}$$

$$\frac{dT}{dt} = -k(T - 20)$$

$$= -k([20 + 160e^{-kt}] - 20)$$

$$= -k(160e^{-kt})$$

$$= -160ke^{-kt}$$

$\therefore T = 20 + 160e^{-kt}$  is a solution of the equation.

ii. At 2:15p.m ( $t = 15$ ):

$$100 = 20 + 160e^{-15k}$$

$$80 = 160e^{-15k}$$

$$e^{-15k} = \frac{1}{2}$$

$$-15k = \ln\left(\frac{1}{2}\right)$$

$$k = -\frac{1}{15} \ln\left(\frac{1}{2}\right)$$

$$= 0.0462098\dots$$

$$\approx 0.0462 \text{ (3 sig. figs)}$$

iii. When  $T = 35$ :

$$35 = 20 + 160e^{-0.046t}$$

$$15 = 160e^{-0.046t}$$

$$e^{-0.046t} = \frac{3}{32}$$

$$-0.046t = \ln\left(\frac{3}{32}\right)$$

$$t = -\frac{1}{0.046} \ln\left(\frac{3}{32}\right)$$

$$= 51.459209\dots \text{min}$$

$$\approx 51 \text{ min}$$

The earliest time Callum can ice the cake is 2:51p.m.

Question 7:

- a) i. Let radius of water surface =  $r$

$$\frac{h}{30} = \frac{r}{24}$$

$$30r = 24h$$

$$r = \frac{4h}{5}$$

$$V = \frac{1}{3}\pi r^2 h$$

$$= \frac{1}{3}\pi \left(\frac{4h}{5}\right)^2 h$$

$$= \frac{1}{3}\pi \times \frac{16h^2}{25} \times h$$

$$= \frac{16\pi h^3}{75}$$

ii.  $\frac{dh}{dt} = 0.5 \text{ cm/min}$

$$\frac{dV}{dh} = \frac{48\pi h^2}{75}$$

$$= \frac{16\pi h^2}{25}$$

$$\frac{dV}{dt} = \frac{dV}{dh} \times \frac{dh}{dt}$$

$$= \frac{16\pi h^2}{25} \times 0.5$$

$$= \frac{8\pi h^2}{25}$$

When  $h = 20$  (i.e. 10 cm from the top):

$$\frac{dV}{dt} = \frac{8\pi \times (20)^2}{25}$$

$$= 128\pi \text{ cm}^3 / \text{min}$$

b) i.  $\frac{CA}{\sin \theta} = \frac{k}{\sin \alpha}$

$$\frac{CB}{\sin(180 - [\alpha + \theta])} = \frac{k}{\sin \alpha}$$

$$CA = \frac{k}{\sin \alpha} \sin \theta$$

$$\frac{CB}{\sin(\alpha + \theta)} = \frac{k}{\sin \alpha}$$

$$CB = \frac{k}{\sin \alpha} \sin(\alpha + \theta)$$

$$l = CA + CB$$

$$= \frac{k}{\sin \alpha} \sin \theta + \frac{k}{\sin \alpha} \sin(\alpha + \theta)$$

$$= \frac{k}{\sin \alpha} [\sin \theta + \sin(\alpha + \theta)]$$

- ii.  $\alpha$  is the angle at the circumference standing on chord  $AB$  and the length of  $AB$ ,  $k$  is a constant.

iii.  $\frac{dl}{d\theta} = \frac{k}{\sin \alpha} [\cos \theta + \cos(\theta + \alpha)]$

When  $\theta = \frac{\pi}{2} - \frac{\alpha}{2}$ :

$$\frac{dl}{d\theta} = \frac{k}{\sin \alpha} \left[ \cos\left(\frac{\pi}{2} - \frac{\alpha}{2}\right) + \cos\left(\frac{\pi}{2} - \frac{\alpha}{2} + \alpha\right) \right]$$

$$= \frac{k}{\sin \alpha} \left[ \sin\left(\frac{\alpha}{2}\right) + \cos\left(\frac{\pi}{2} - \left\{\frac{\alpha}{2}\right\}\right) \right]$$

$$= \frac{k}{\sin \alpha} \left[ \sin\left(\frac{\alpha}{2}\right) + \sin\left(-\frac{\alpha}{2}\right) \right]$$

$$= \frac{k}{\sin \alpha} \left[ \sin \frac{\alpha}{2} - \sin \frac{\alpha}{2} \right]$$

$$= \frac{k}{\sin \alpha} \times 0$$

$$= 0$$